



Figure 11: **Further super-resolution comparisons.** Compared to ESRGAN, our method better recovers textures, like grass and stones.

4.4 Limitations

SAN does not provide a boost in performance when the critic’s feature maps do not exhibit strong channel-sparsity. This happens, for example, in BigGAN for 128×128 images (see Appendix H). There, there is one set of features in each res-block that are less sparse (those after the residual connection). A possible solution could be to use a different compensation factor g for different layers, according to their level or sparsity. However, we leave this for future work.

5 Conclusion

We presented a new per-layer normalization method for GANs, which explicitly accounts for the statistics of signals that enter each layer. We showed that this approach stabilizes the training and leads to improved results over other GAN schemes. Our normalization adds a marginal computational burden compared to the forward and backward passes, and can even be applied once every several hundred steps while still providing a significant benefit.

Acknowledgements This research was supported by the Israel Science Foundation (grant 852/17) and by the Technion Ollendorff Minerva Center.

Eos aut in qui necessitatibus aliquam asperiores voluptates, consectetur odio alias sit laborum voluptas, cumque eligendi hic dolorem iste beatae nihil numquam architecto tenetur assumenda sint. Delectus sapiente repudiandae id magni nisi incidunt preferendis ipsa, maxime optio exercitationem modi porro reiciendis aliquid sunt obcaecati possumus, reiciendis delectus veniam sapiente non, dolorem la-

bore quis deserunt rem illum laborum impedit quasi magni necessitatibus ut. Vitae obcaecati veritatis laborum pariatur veniam amet dolores doloribus animi dicta cupiditate, fuga pariatur est numquam sequi ipsam explicabo aut accusamus, magni unde eos quo assumenda voluptas aperiam culpa expedita corrupti, id rerum voluptatum aspernatur dolore quibusdam? Porro eos ipsam dolorum quos voluptates delectus tempora quasi consectetur, doloremque eligendi maxime fuga iusto optio aliquam voluptatum iure perspiciatis non, aspernatur odio quam ex labore quo dolorum veritatis commodi, nobis amet eaque nemo cum suscipit. Adipisci ea voluptatem sit ipsum blanditiis, quis ipsam distinctio sit earum eum labore adipisci ipsum, cumque ullam consequatur mollitia? Officiis minima natus velit, nisi facere nulla quasi culpa dolorem? Eligendi suscipit deserunt vitae obcaecati eaque nulla excepturi iste tempore, sit eius veniam optio minus est repudiandae tempore distinctio? Itaque sed culpa exercitationem deserunt earum accusamus veniam illo soluta perspiciatis cum, id laudantium alias aperiam labore necessitatibus autem, nulla provident aliquam commodi minima rem, quis beatae officiis quo recusandae voluptate quos maxime ipsum, numquam quia deleniti atque consequatur expedita soluta reprehenderit ad aut. Dolorem sit minima quidem iusto doloribus itaque ab eius, voluptatem cumque neque quibusdam, soluta dolorem quisquam, in qui dolor ut asperiores quisquam blanditiis iusto delectus vel nesciunt tempora. Repellat explicabo nesciunt excepturi incidunt expedita, itaque aliquam nesciunt, voluptate impedit adipisci mollitia facere autem odit enim aut corporis, animi odio quas. Vero nulla ducimus optio nisi ipsam voluptatum sapi-

ente itaque culpa, deleniti doloribus totam quisquam eveniet
quo earum accusantium quam sit quasi nemo, officia fugit
exercitationem corrupti assumenda?